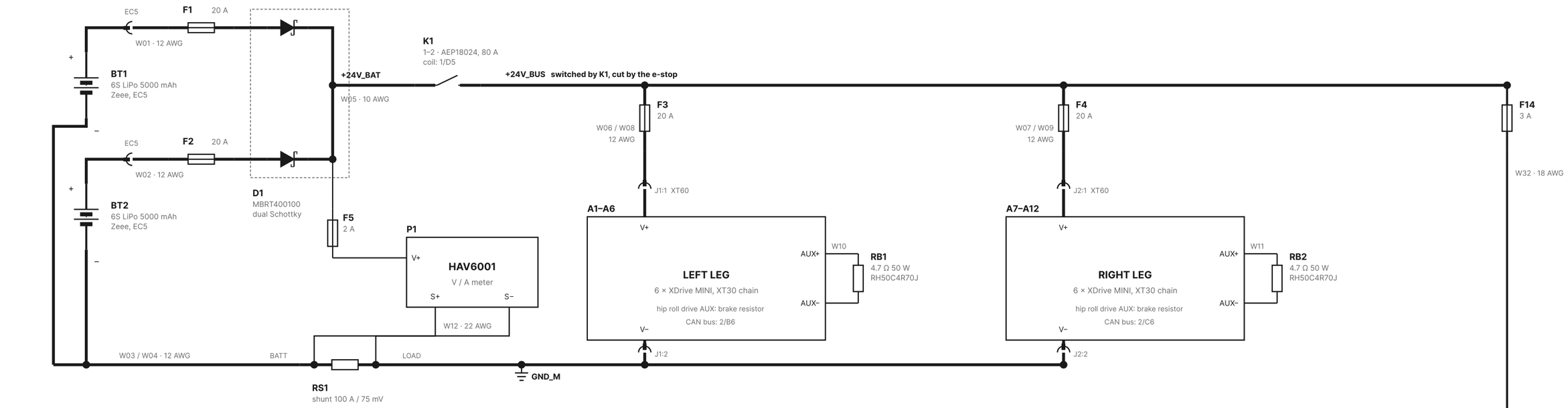
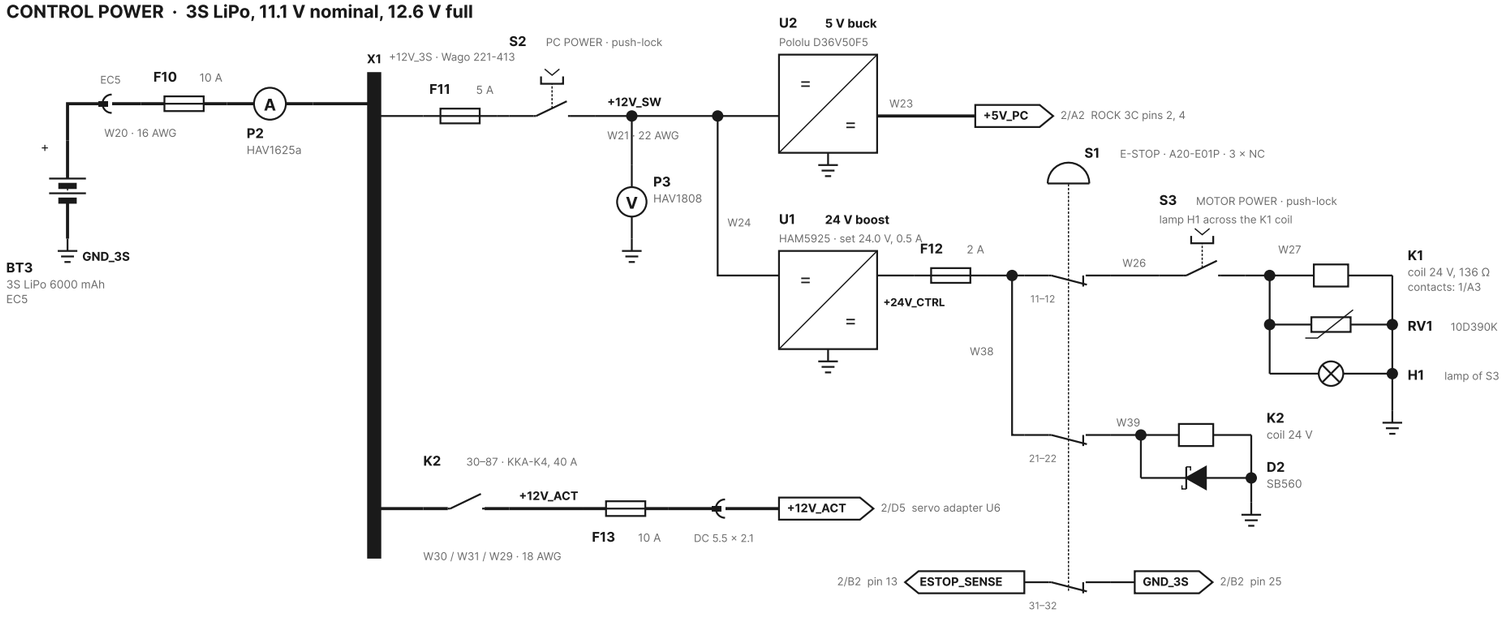
MOTOR POWER · 2 × 6S LiPo, 22.2 V nominal, 25.2 V full

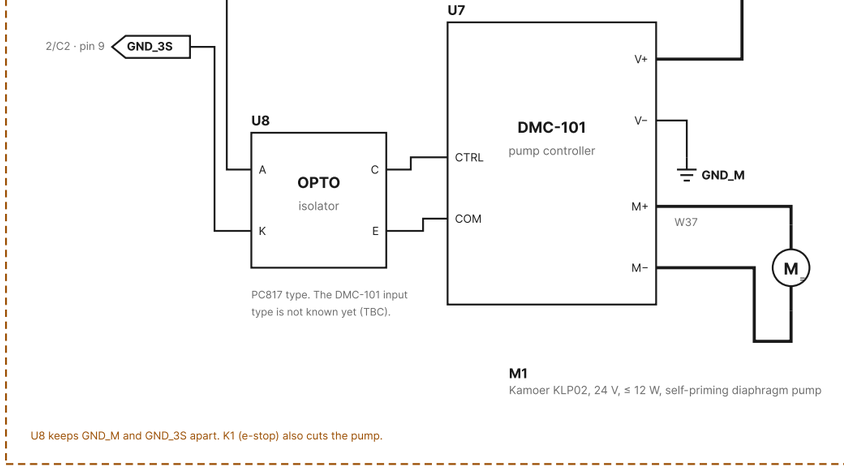


CONTROL POWER · 3S LiPo, 11.1 V nominal, 12.6 V full



WATER PUMP · 24 V

PROPOSED – CONFIRM WITH THE DMC-101 DATASHEET



U8 keeps GND_M and GND_3S apart. K1 (e-stop) also cuts the pump.

NOTES

1. Contacts are drawn at rest: e-stop released, push-lock switches open, relays not energised.
2. Fuses: mini blade, 32 V DC. F1–F4 in MTA 0100335 holders, the others in Coms BE151 holders.
3. Wire numbers (W01 …) and gauges refer to docs/hardware/harness.csv. Thick lines: 10–12 AWG.
4. GND_M and GND_3S share no wire. The CAN adapters are isolated (sheet 2).
5. Set U1 to 24.0 V and its current limit to 0.5 A before you connect the K1 and K2 coils.
6. RB1, RB2: set brake_resistance = 4.75 Ω on the two hip-roll drives only (electrical.md, 3.4).

Rev	Date	Change
A	2026-10-15	First wiring plan (final design F1, 12 V pump)
B	2026-10-22	Final design F2: 24 V pump on the motor bus, new drawing set

SOBANG – firefighting humanoid on Berkeley Humanoid Lite legs

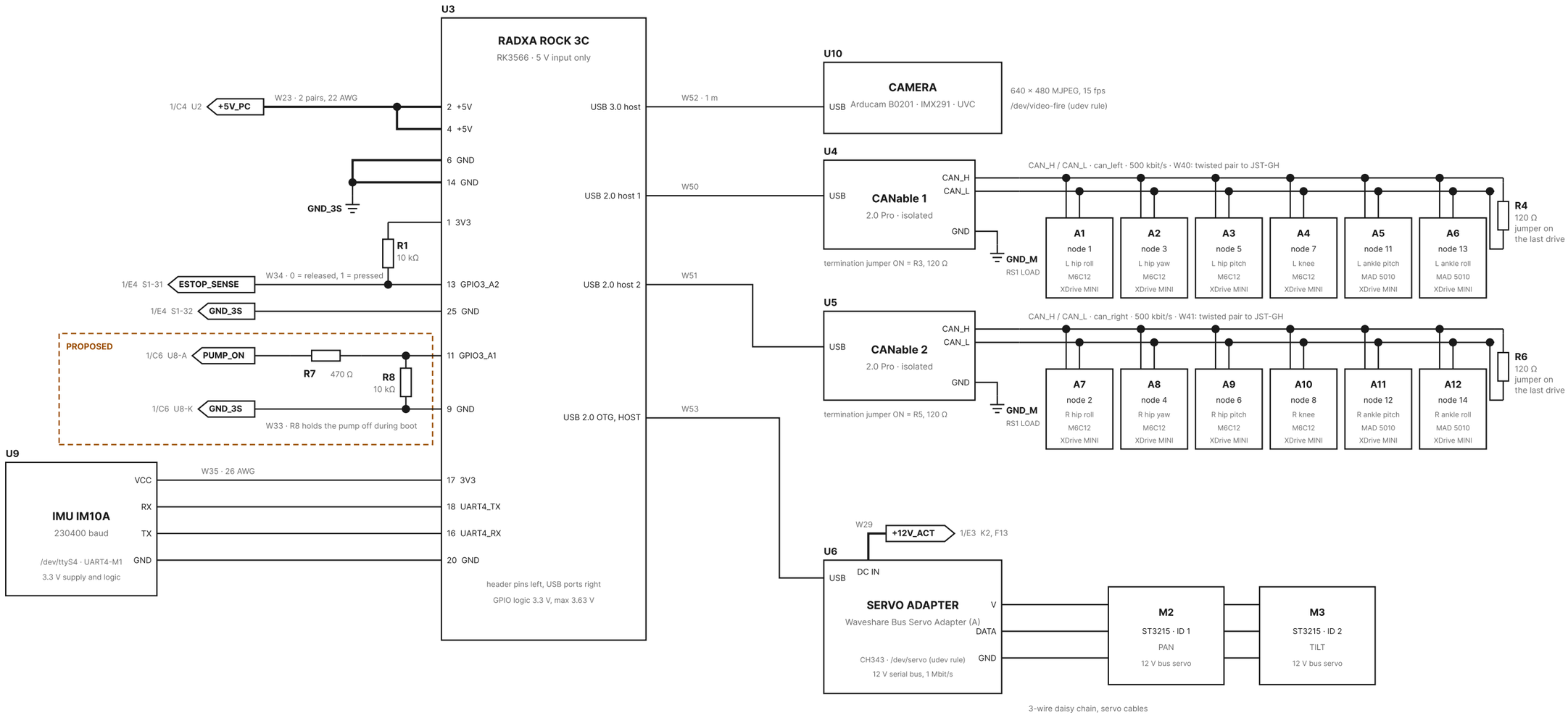
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Power distribution

Drawing	Rev
SOB-E-01	B
Sheet	Date · Size
1 of 2	2026-10-22 · A3

Status: preliminary, not built. Sources: docs/hardware/electrical.md, harness.csv

ROBOT COMPUTER, CAN BUSES, SENSORS AND PAN-TILT



NOTES

1. Node ids follow legs/robot.py (JOINTS, BHL numbering). If the drives get other ids, change that table.
2. The order of the drives along each chain is not fixed. Terminate only the last drive of each chain.
3. Each CANable joins its CAN GND to GND_M at the RS1 LOAD bolt only (W40, W41).
4. Do not plug a USB-C supply into the ROCK 3C while the header gives 5 V: both feed the same rail.
5. Do not use header pins 8 and 10: they carry the boot console. Enable UART4-M1 with rsetup.
6. gpiochip numbers are not verified: check GPIO3_A1 and GPIO3_A2 with gpiodetect on the board.

Rev	Date	Change
A	2026-10-15	First wiring plan (final design F1, 12 V pump)
B	2026-10-22	Final design F2: 24 V pump on the motor bus, new drawing set

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Control and communication	Drawing **SOB-E-02**	Rev **B**
Status: preliminary, not built. Sources: docs/hardware/electrical.md, harness.csv	Sheet **2 of 2**	Date · Size **2026-10-22 · A3**